

What Is a False Negative Worth?

Three Ways an Asymmetric-Cost Analysis Fails Review, and What Survives It

Kartik Thakore
Hammer Labs
kartik@hammer.ai

September 9, 2026

Abstract

Cost-sensitive learning asks for one number: how many false alarms a missed positive is worth. In applied work that number is almost always asserted, and the analysis built on it then has to survive somebody whose job is to audit the return. This paper is written for the person doing the surviving.

We report three failure modes, each measured on a corpus of 156,732 SEC filings labelled for later retraction, plus two public control datasets. **First, the ratio decides the result and is usually borrowed.** Three independent feature spaces agree that our models beat the trivial baseline below $c \approx 5.6$ and lose above it, and the constant our own work had inherited turned out to be a model-selection key from an unrelated task in another repository. **Second, the baseline may be one nobody can staff.** Under $\text{FN} \cdot c + \text{FP}$ always-fire pays only the per-review price at any volume, so it wins whenever the base rate clears $1/(c + 1)$, and a benchmark scored that way reports the base rate while appearing to report discrimination. **Third, and largest, the decision threshold is often selected in-sample.** That inflates test cost by up to $5.5\times$, is governed by how completely the model memorises its training set rather than by p/n or model quality, and is therefore worst for the strongest model on the slide. Calibration provably cannot repair it; out-of-fold selection recovers a median 95% of the loss for K extra fits and no held-out data. A rolling-origin temporal split leaves the effect the same size, and moves the oracle threshold not at all: a forward split does not reduce what a model can achieve, it taxes every strategy that has to carry a threshold across it.

We give a procedure for deriving the cost ratio from published labor rates rather than asserting it, and report where that procedure breaks: the derived ratio depends on the seat the model is deployed in, so a task specification that does not name a seat does not determine a ratio. Where the seat is genuinely ambiguous, evaluating at a stated review capacity is defensible in a way a single asserted ratio is not, because $\text{precision}@k$ requires no cost ratio at all. We close with the questions a reviewer will ask and what each of them requires you to have already measured.

1 Introduction

A practitioner building a detector for a business process is handed an asymmetric problem. A missed fraud, a missed defect, a missed restatement is worse than a false alarm. The standard response is a cost-sensitive loss,

$$\text{cost} = \text{FN} \cdot c + \text{FP},$$

and the standard practice is to pick c by assertion: a round number, a number from a paper in the same domain, a number a stakeholder offered in a meeting.

Then somebody audits the return. The analysis has to hold up in a room where the question is not whether the model is interesting but where each number came from, and a figure whose provenance is “it is what people use” does not survive that room. This paper is written for that seat: the analyst, data science lead or consultant who did the work and now has to defend it, rather than for the person doing the auditing and rather than for a methods audience.

What that reader needs is not a finding. It is to know which parts of the analysis will be attacked, what each attack costs if it lands, and what has to have been measured beforehand for the answer to be more than an opinion. So the paper is organised as three failure modes, in increasing order of how much they cost and how hard they are to see.

Contributions.

- (1) **We show the ratio decides the answer, and quantify where the decision flips.** On our corpus the crossover is $c \in [5.5, 5.75]$ across three unrelated feature spaces. Below it every model beats the trivial baseline; above it every model loses. Nothing about the models changes.
- (2) **We give a procedure for grounding c in labor hours** and run it. This is, to our knowledge, the first attempt in this literature to derive an asymmetric cost ratio from published compensation and fee data rather than assert it.
- (3) **We identify a failure of that procedure that is not a data gap.** The derived ratio depends on the seat the model is deployed in, and a task specification that does not name a seat does not determine a ratio. We argue this is common and under-recognised.
- (4) **We report a threshold-selection artefact** worth up to $5.5\times$ in test cost, establish its mechanism, show that the obvious remedy provably cannot work, and benchmark eight remedies that can.

2 The ratio is asserted, not measured

Cost-sensitive learning has a mature theory [1, 2] and mature machinery: MetaCost, class weighting, instance-dependent variants [3]. The theory takes the cost matrix as given. The applied literature then supplies it by assertion. Surveying recent financial-misstatement work, we found ratios of 6.46:1 and similar figures stated without derivation.

Our own case is worse and is instructive. Our repository’s constant traced to a single table row in a different repository:

```
| selection | val flip-pair accuracy | val asymmetric key (missed×10 + false)
|
```

It was the metric used to select a training epoch, for a different model, on 178 synthesised rows, answering a different question. It was never a claim about our domain. Every result we had published was reported against it as though it described the world.

An unmeasured constant is at least a guess about the right domain. This one had crossed a repository, a corpus and a decision problem. We suspect this is not rare.

c	XBRL arm	narrative arm	embedding arm
3	0.652	0.576	0.661
5	0.903	0.882	0.916
10	1.572	1.407	1.273

Table 1: Arm cost over the best floor. Below 1.0 the model wins. Same models, same rows, same split; only c moves. On a quarter-point grid the crossover is 5.5, 5.75 and 5.5.

3 The ratio decides the result

Setup. 156,732 SEC filings from 5,085 companies, 2004–2024. Labels come from 8-K Item 4.02, the item a registrant files to state that previously issued financials should no longer be relied upon; the 8-K names the filings it retracts, so labels are mechanical rather than annotated. Two views per filing: narrative section text, and XBRL facts. Splits are by company throughout, never by row, because filings from one company are not independent.

We score each arm against *the best floor*, which is the cheaper of always-fire (cost = N) and never-fire (cost = $P \cdot c$) at that ratio. Which floor wins is itself a function of c , so comparing against a fixed one confounds two effects.

Table 1 is the whole argument of Section 2 in one place. Three feature spaces with nothing in common agree on the crossover to within a quarter of a point, and the constant we had inherited sits on the far side of it.

3.1 What the linear form hides

The form itself has a defect independent of the constant. Under $FN \cdot c + FP$, always-fire reviews 100% of cases and pays only the per-review price. It is therefore purchasable at any scale, and it wins whenever the base rate clears $1/(c + 1)$.

No ranking model can beat a strategy that takes every positive by taking every row.

A corpus scored this way is answering “is the base rate high” while appearing to answer “does the model discriminate.” Writing the cost per row with base rate π ,

$$\frac{\text{cost}}{n} = c\pi(1 - \text{TPR}) + (1 - \pi)\text{FPR},$$

always-fire is the ROC point $(1, 1)$, and a model beats it exactly when

$$c < \underbrace{\frac{1 - \pi}{\pi}}_{\text{odds of a negative}} \cdot \underbrace{\max_{\text{ROC}} \frac{1 - \text{FPR}}{1 - \text{TPR}}}_{\text{base-rate free}}.$$

This is a restatement of iso-performance analysis [2] and is not new geometry. What is worth saying to practitioners is the consequence: the contest is decided by the steepest line from the *top-right* corner down to the ROC curve, and a model can have excellent AUC while being flat exactly there. Reading a ROC curve from $(0, 0)$ tells you nothing about it.

detection rate d	0.25 h review	0.5 h	1 h	4 h	8 h
0.10	304	152	76	19	10
0.25	760	380	190	48	24
0.50	1520	760	380	95	48
1.00	3040	1520	760	190	95

Table 2: Implied c , from $c = d \cdot H_{\text{miss}}/H_{\text{review}}$ with $H_{\text{miss}} = 760$ professional hours. One cell in twenty reaches the constant our work had been using, and it is the corner: a review that finds the problem one time in ten, costing a full working day.

4 Grounding the ratio in labor hours

If the ratio must be supplied, supply it from something a person can be asked about. Both halves of the following are answerable by an operations lead:

$$c = \frac{\text{hours a miss eventually costs}}{\text{hours one review costs}}.$$

The recipe.

- 1. Name the seat.** Who acts on the prediction, and what do they do differently when it fires? This step is not optional; Section 4.1 is what happens when it is skipped.
- 2. Price the review.** Hours per case at that seat, times a loaded rate. Triage and full review differ by an order of magnitude and the answer moves with them.
- 3. Price the miss, in the same units.** Published fee premia, remediation studies, incident post-mortems, and internal time-tracking are all admissible. State which.
- 4. Apply the detection discount.** A flagged case only avoids the downstream cost if the review actually finds the problem. Multiply by that rate, call it d , and be explicit that you are guessing at it if you are.
- 5. Report the grid, not the point.** Steps 2 to 4 each carry uncertainty. A table across their plausible ranges is a result; a single number is a claim you cannot support.

Worked example. An interim restatement carries a mean audit fee premium of \$191,000, the gap between a restating client’s \$1.583M mean fee and the \$1.392M it would otherwise have paid [5]. At the published US blended range of \$150 to \$450 per hour that is 424 to 1,273 professional hours. It is the auditor alone; internal controller, legal and consultant hours do not appear in any standardised disclosure, so this is a lower bound rather than an estimate. Taking $H_{\text{miss}} = 760$ hours (the midpoint rate):

Every cell but one is between two and three hundred times the asserted constant, and the crossover from Table 1 is 5.5. The practical reading is uncomfortable and worth stating plainly: **at the ratio the domain actually implies, flagging everything is correct on cost, and no model in this study was ever going to win that argument.**

4.1 The seat problem

The 760 hours belongs to a seat our corpus does not occupy, and noticing this is the most transferable result in the paper.

Our task is post-filing: given a document already filed, predict whether it will later be retracted. Flagging it *does not prevent the restatement*. The error is in the numbers and the remediation happens either way. What early detection buys depends on who is asking:

- **A preparer or auditor, before filing.** Catching it here does avoid the cost, and 760 hours is the right numerator. But this seat cannot be our corpus, because the label only exists after the fact.
- **An investor.** A miss is holding a position through an announcement effect of roughly -9.2% [4]. The cost scales with position size, so a unitless ratio against “one analyst hour” is not the right shape for the question.
- **A regulator or reviewer triaging filings.** The remediation happens regardless. What is bought is *earlier*, valued in months rather than in hours avoided.

A task specification that does not name a seat does not determine a cost ratio.

This is not a gap that better data fills. It is a statement about what the task is. We suspect many benchmark tasks with asserted cost matrices are in this position, and that the asserted constant is doing the work of a missing problem statement.

The way out. Where the seat is genuinely ambiguous, evaluate at a stated *capacity* instead. A reviewer has k slots, so the decision is a queue and not a label, and always-fire leaves the choice set because it cannot say which k to open. $\text{precision}@k$ requires no cost ratio at all [6], only a number every operations team already knows.

The change is not cosmetic. Under a top-1% budget on the 2021 window, our narrative arm’s top 11 ranked filings are all later-retracted, and so are its top 22: it reaches the $1/\pi$ ceiling exactly. Under $\text{FN} \cdot 10 + \text{FP}$ the same arm on the same scores reads as 41% worse than useless. Both statements are true, and only one describes something a team could staff.

5 The threshold was a larger error than the ratio

While measuring the above we found a defect in our own pipeline that dominated it, and that we believe is widespread because it is invisible.

A score is not a decision. Somewhere a threshold is chosen. Our code, and much applied code, chose it by sweeping the model’s scores *on the rows the model was fitted on*.

5.1 Measurement

Table 3 benchmarks eight threshold-placement strategies over 180 cells: $5 \text{ datasets} \times 3 \text{ seeds} \times 3 \text{ models} \times 4 \text{ cost ratios}$. Datasets are our SEC arms plus two public controls, **bank-marketing** and **creditcard**, both from OpenML. Models are a random forest, histogram gradient boosting, and L_2 logistic regression.

In-sample selection is dominated on every column. Its worst case is 4.581, its regret is $4.7 \times$ the next worst strategy, and it is worse than doing nothing in 39% of cells.

5.2 It is worst for the models you would deploy

The natural objection is that a weak model produced this. The opposite holds.

strategy	mean	median	worst	mean rank	regret	>floor
S4 held-out validation	0.695	0.670	1.242	3.09	0.033	13%
S6 analytic + OOF isotonic	0.765	0.689	1.968	4.61	0.103	23%
S1 out-of-fold (5-fold)	0.766	0.702	1.966	3.49	0.105	23%
S3 OOF, bootstrap median	0.768	0.682	1.966	4.08	0.106	23%
S2 OOF, 3 repeats averaged	0.778	0.697	2.200	4.03	0.117	23%
S7 firing-rate match	0.809	0.696	2.290	4.83	0.147	25%
S5 analytic $1/(1+c)$, raw	0.851	0.696	3.551	5.14	0.189	23%
S0 in-sample	1.156	0.910	4.581	6.72	0.494	39%

Table 3: Eight threshold strategies, 180 cells, cost over the best floor. *regret* is the mean gap to an oracle threshold chosen on test, which is the only column that isolates the line from the model. *>floor* is how often the strategy did worse than flagging everything.

dataset	model	p/n	train AUC	inflation @ $c=20$	val/floor
bank-marketing	L_2 logistic	0.002	0.909	1.00–1.02	0.36
bank-marketing	random forest	0.002	1.000	4.90–5.51	0.29
SEC narrative	gradient boost	1.385	1.000	2.62–3.76	0.90
SEC embedding	L_2 logistic	0.354	0.908	0.96–1.14	0.96

Table 4: Inflation is the cost of a train-selected threshold over a validation-selected one, on identical test rows, three seeds. Rows one and two are the same dataset, the same 22,605 training rows, the same 51 features and the same splits.

Across all cells, every model reaching a training AUC of exactly 1.000 shows the artefact (inflation 1.27 to 5.51); no model at 0.91 does (1.00 to 1.14). The governing variable is **train-set memorisation**, not p/n and not model quality. Our own first analysis attributed it to p/n ; that was an artefact of having only one learner, and Table 4 refutes it.

5.3 Mechanism: firing-rate drift

A threshold is chosen so that some fraction of training rows fires. Applied to new rows it fires on a different fraction, and that drift is what the cost pays for.

model (bank, $c=20$)	fires on train	fires on test	inflation
random forest	0.117	0.050	4.90
L_2 logistic	0.410	0.417	1.00

The forest’s threshold was set to fire on one case in nine and fires on one in twenty. AUC gap does *not* predict the magnitude and drift does: AUC gap is a mismatch in *ranking*, while a threshold is exposed to a mismatch in the score *distribution* at the operating point. This is why an uncalibrated forest, whose training scores pile at 0 and 1, is the worst offender in the grid despite ranking well.

5.4 Calibration cannot fix it, and this is a proof rather than a measurement

The obvious remedy is calibration. It does nothing, in all 84 cells we ran, identically to four decimal places.

A calibrator is a monotone map. For strictly increasing g , $\{s \geq t\} = \{g(s) \geq g(t)\}$, so the family of achievable confusion matrices is unchanged and so is its cost-optimal member. Isotonic regression is non-decreasing rather than strictly increasing, so ties could in principle break differently; across 84 cells they never did.

Calibration fixes what a score *means*. This is a problem about which rows a score *selects*.

Where calibration does earn its keep. The analytic Bayes rule for this loss is to fire when $p > 1/(1+c)$, since firing costs $(1-p)$ and not firing costs pc . That rule reads the score’s *value* rather than its rank, so a calibrator is exactly what it needs. Applied to out-of-fold isotonic scores it reproduces the empirical out-of-fold sweep in all 180 cells (S6 versus S1 in Table 3: 0.765 against 0.766, worst case 1.968 against 1.966). One tool, inert where the procedure reads ranks and load-bearing where it reads values.

Applied to *raw* model output it fails on saturation, not on miscalibration: boosting’s firing rate moves only from 0.289 to 0.340 as c goes from 3 to 20, when the sweep wants 0.637. Scores piled at 0 and 1 mean that dragging the threshold to 0.048 crosses almost no rows, so the rule cannot respond to c at all.

5.5 A forward split leaves it the same size, and does not move the oracle

Every result above splits by company. A reviewer of a financial prediction will not accept that, because a deployment only ever holds the past, so we ran a rolling origin with a gap block: train on filings from years $[Y-4, Y]$, select on $Y+1$, report on $Y+2$, at five origins. **Each origin is run twice, once forward in time and once by company over the identical pooled rows at identical block sizes**, because a temporal number on its own cannot say whether anything changed. 240 cells.

S0/S4	$c=3$	$c=5$	$c=10$	$c=20$	all
temporal	1.076	1.031	1.180	1.663	1.238
company	1.026	0.997	1.163	1.625	1.203
paired difference	+0.050	+0.034	+0.017	+0.038	+0.035

The in-sample penalty is larger under the temporal split in 69 of 120 paired cells, which is a coin flip, and dropping any single origin moves the difference between +0.009 and +0.067 on a baseline of 1.20. A prior that distribution shift multiplies the memorisation effect predicts something on the scale of the effect itself, which is 20 to 66 percent; six percent is not that. **We report it as the same size and give no growth factor.**

What a forward split does instead is more interesting, and it is visible in the regret column:

mean regret against the oracle	temporal	company	ratio
S0 in-sample	0.320	0.231	1.38
S1 out-of-fold	0.102	0.044	2.30
S4 validation	0.069	0.033	2.08
oracle, absolute cost	0.901	0.890	1.01

The oracle does not move. The best threshold obtainable on the test block costs the same whether that block was reached by calendar or by company, so a forward split does not

dataset	train n	in-sample	out-of-fold	validation
bank-marketing	22,605	0.672	0.399	0.399
bank-marketing (cap)	300	1.201	0.699	0.656
SEC XBRL	2,167	1.178	0.874	0.772
SEC narrative	2,167	1.398	0.912	0.795

Table 5: Mean cost over floor. At 22,605 training rows out-of-fold is free and exactly as good as spending a quarter of the data. At 2,167 it is not, and the tail is louder than the mean: worst case 1.966 against 1.123.

reduce what the model can achieve. It taxes every strategy that has to *carry* a threshold across the split, and the oracle carries none. The careful strategies pay proportionally more: out-of-fold and held-out selection roughly double their regret, while the in-sample threshold, already bad, worsens by 38 percent.

For the reader defending an analysis, that is the useful form of the result. A reviewer asking “does this hold up out of time” is asking two questions at once, and they have different answers: the model’s achievable performance is unchanged, and the cost of transferring an operating point roughly doubles.

5.6 What does work is a size rule

Two natural refinements that attack *variance* buy nothing, which we verified: averaging out-of-fold scores over three repeats and taking a bootstrap-median threshold both move the mean by less than 0.03, and the repeated variant is marginally worse in the tail.

We do not have an explanation for the residual gap at small n , and record a rejected one. If it were simply that out-of-fold models are fitted on $(K-1)/K$ of an already small set, raising K would close it. Sweeping $K \in \{2, 5, 10, 20\}$ does not: the out-of-fold minus validation gap runs 0.078, 0.041, 0.110, 0.112, which is non-monotone and worst at the largest K . That sweep is not a clean comparison, since the $K=5$ cells carry more seeds than the others, but nothing in it supports the mechanism, so the residual stays unexplained.

6 What a reviewer will ask, and what it requires you to have measured

Each of these is a question that arrives in the room, the answer that survives it, and the thing that must already have been measured for the answer to be more than an opinion.

1. **“Where did that cost ratio come from?”** Derive it from hours, and report the grid rather than the point: hours a miss eventually costs, over hours one review costs. Both halves are answerable by an operations lead. *Requires:* your own review time per case, and a published or internal figure for downstream cleanup. Neither is in a vendor’s deck.
2. **“Who is this cost ratio for?”** Name the seat before naming the number. A preparer, an investor and a regulator have three different numerators for the same prediction, and one of them does not reduce to a ratio at all. *Requires:* a sentence saying who acts on the prediction and what they do differently when it fires. If you cannot write it, you do not have a cost ratio, you have a missing problem statement.

3. **“So should we just review everything?”** Sometimes yes, and saying so is what makes the rest credible. At the ratio our domain implies, always-fire is correct on cost and loses only because it cannot be staffed. *Requires:* the review capacity, stated as a number. Then report $\text{precision}@k$ against it, which needs no cost ratio.
4. **“Why is production worse than the pilot?”** Because the threshold was chosen in-sample. Choose it out of fold, or on a held-out block when the training set is small, and say which in the write-up. *Requires:* nothing you do not already have. K extra fits.
5. **“Does this hold up out of time?”** Two questions with different answers. The model’s achievable cost is unchanged under a forward split; the cost of carrying an operating point across one roughly doubles (§5.5). *Requires:* a rolling origin, and a same-size company split beside it, because a temporal number alone cannot say whether anything moved.
6. **“What is the basis of this number?”** Tag every figure as measured, cited or assumed, and let anything computed from a mix inherit the weakest. An annual saving computed from a measured catch rate and an assumed hourly rate is an assumption, and a write-up that says so survives a reviewer who was going to find out anyway.

What not to do. Do not reach for calibration to fix a threshold; it is provably inert unless the decision rule reads score values rather than ranks. Do not report a single-ratio asymmetric cost as a finding: “the model does not beat the baseline” is not a result, while “the model does not beat the baseline above 5.75:1” is a different one that a reader can act on. Do not spend effort on variance reduction in threshold selection; the residual is not variance, and we could not establish what it is.

7 Limitations

Three seeds on most cells, eight on the two headline cells, and two on the temporal grid; the company split is primary and the temporal one is Section 5.5. The hours derivation rests on the auditor’s side of the cost, which is published, and treats the company’s internal effort as unmeasured rather than zero, so Table 2 is a lower bound on H_{miss} and its detection-rate axis is a stated guess rather than a measurement. Two of the three model families are tree ensembles, so the saturation finding may not transfer to deep models whose score distributions differ. The oracle threshold fits one free parameter to the test set and is reported as a bound rather than an achievable number.

8 Related work

Cost-sensitive learning theory [1] and ROC iso-performance analysis [2] give the geometry we re-derive in Section 3; our contribution there is the empirical crossover and the observation that it sits inside the range of asserted constants. Instance-dependent cost [3] motivated our firm-size weighting experiment, which was a null. Consequentialist critique [6] argue for $\text{precision}@k$ under known capacity, which we adopt and to which we add a variance argument: a fixed budget has no threshold to estimate. Threshold selection is treated as a tuning detail in most applied work; we are not aware of prior work quantifying the in-sample selection penalty as a function of memorisation.

References

- [1] C. Elkan. The foundations of cost-sensitive learning. *IJCAI*, 2001.
- [2] F. Provost and T. Fawcett. Robust classification for imprecise environments. *Machine Learning*, 42(3):203–231, 2001.
- [3] A. C. Bahnsen, D. Aouada, and B. Ottersten. Example-dependent cost-sensitive logistic regression for credit scoring. *ICMLA*, 2014.
- [4] U.S. General Accounting Office. Financial statement restatements: trends, market impacts, regulatory responses, and remaining challenges. GAO-03-138, 2002.
- [5] Interim restatements and the audit engagement. *Journal of Accounting and Public Policy*, 2025.
- [6] A consequentialist critique of binary classification evaluation: theory, practice, and tools. arXiv:2504.04528, 2025.